Triple-Higgs production as a probe of the Higgs potential

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Abstract

We study triple-Higgs production in the κ-framework as a direct probe of the Higgs potential. We derive compact parametrisations of the inclusive gluon-fusion production cross section that capture the dependence on the Higgs trilinear and quartic self-couplings together with anomalous top-Higgs interactions. These results provide a practical framework for interpreting current and future LHC measurements. We use the parametrisations to assess the sensitivity of LHC Run 2 and HL-LHC data to Higgs self-couplings and show that differential kinematic distributions can provide additional discrimination between scenarios with similar inclusive production rates.

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1 Introduction

The discovery of the Higgs boson in 2012 [1, 2] completed the particle content of the Standard Model (SM) and established the mechanism of electroweak (EW) symmetry breaking. Since then, a major objective of the LHC physics programme has been to determine the properties of the Higgs boson with increasing precision. While its couplings to EW gauge bosons and third-generation fermions are already constrained at the few- to ten-percent level, its self-interactions remain largely unexplored. Measuring the Higgs self-couplings is therefore one of the most important outstanding goals of collider physics, since they directly probe the shape of the Higgs potential and the dynamics of EW symmetry breaking.

In the SM, the Higgs potential is determined by a single quartic coupling $\lambda = m_h^2/(2v^2)$. After EW symmetry breaking, it predicts unique values for the Higgs trilinear and quartic self-couplings,

$$\lambda_3^{\rm SM} = \lambda\,, \qquad \lambda_4^{\rm SM} = \frac{\lambda}{4}\,, \tag{1}$$

once the Higgs boson mass $m_h \simeq 125\,\text{GeV}$ and vacuum expectation value $v \simeq 246\,\text{GeV}$ are fixed. Any deviation from these predictions would provide evidence for physics beyond the SM, such as extended scalar sectors, composite Higgs models, or mechanisms for EW baryogenesis. Reviews of such beyond-the-SM (BSM) scenarios can be found, for example, in Refs. [3, 4].

At the LHC, double-Higgs production provides the primary direct probe of the Higgs trilinear self-coupling, while loop-induced effects in single-Higgs processes yield complementary constraints [5–13]. These measurements are typically interpreted in the κ-framework [14–16], which parametrises deviations from the SM through

$$\kappa_3 = \frac{\lambda_3}{\lambda_3^{\rm SM}}\,, \tag{2}$$

assuming all other couplings take their SM values. Using the full $\sqrt{s} = 13\,\text{TeV}$ LHC Run 2 data set, the ATLAS and CMS collaborations obtain the combined constraint

$$-0.71 < \kappa_3 < 6.1 \quad (\text{ATLAS} + \text{CMS}) \tag{3}$$

at the 95% confidence level (CL) [17]. The HL-LHC, with an integrated luminosity of $6\,\text{ab}^{-1}$, is expected to improve this sensitivity to

$$0.5 < \kappa_3 < 1.7 \quad (\text{HL-LHC})\,. \tag{4}$$

Constraining the quartic Higgs self-coupling is considerably more challenging. It is conventionally parametrised as

$$\kappa_4 = \frac{\lambda_4}{\lambda_4^{\rm SM}}\,, \tag{5}$$

and triple-Higgs production provides its only direct probe at hadron colliders. However, the tiny SM cross section of approximately 100 ab at $\sqrt{s} = 14\,\text{TeV}$, calculated at next-to-next-to-leading order (NNLO) in QCD [18], renders such measurements experimentally very challenging. Current ATLAS [19] and CMS [20] searches yield

$$-230 < \kappa_4 < 240 \quad (\text{ATLAS})\,, \qquad -190 < \kappa_4 < 190 \quad (\text{CMS})\,, \tag{6}$$

at the 95% CL, assuming all other couplings are SM-like, in particular $\kappa_3 = 1$. A recent HL-LHC study [21] projects these bounds to improve to

$$-81 < \kappa_4 < 89 \quad (\text{HL-LHC}) \tag{7}$$

with an integrated luminosity of $6\,\text{ab}^{-1}$. In addition to triple-Higgs production, double-Higgs production is also sensitive to κ_4 through two-loop EW corrections [21–24], although these constraints are generally weaker.

In this note, we present a comprehensive study of triple-Higgs production within the κ-framework. We derive compact parametrisations of the inclusive gluon-fusion production cross section that capture its dependence on the Higgs trilinear and quartic self-couplings, the top-quark Yukawa coupling modifier κ_t, and the two-top-two-Higgs interaction κ_{2t}. These expressions provide practical tools for interpreting current and future LHC measurements. Using them, we investigate the sensitivity of current and projected LHC data to the Higgs self-couplings and study the impact of correlated modifications of κ_t and κ_{2t}. We further demonstrate that differential kinematic distributions retain significant sensitivity to the Higgs potential even in scenarios with nearly degenerate inclusive cross sections, highlighting the importance of shape information in future triple-Higgs searches.

2 Triple-Higgs production in the κ-framework

Triple-Higgs production at hadron colliders has been the subject of extensive theoretical investigation over the past two decades [18, 21, 22, 25–41]. Within the SM, the gluon-fusion process $gg \to 3h$ is dominated by the one-loop topologies shown in the upper and middle rows of Figure 1. In the figure, green (blue) boxes denote insertions of the Higgs trilinear (quartic) self-coupling, while orange circles represent the top-quark Yukawa coupling. The two diagrams in the lower row illustrate representative BSM contributions involving a contact interaction between two top quarks and two Higgs bosons. In the SM, bottom-quark loops contribute only about −0.2% to the $gg \to 3h$ production cross section at LHC energies [39]. We therefore neglect all light-quark contributions throughout the remainder of this work.

The κ-formalism Lagrangian, which parametrises the interaction vertices highlighted in Figure 1, takes the form

$$\mathcal{L}_\kappa \supset -\kappa_3 v \lambda h^3 - \kappa_4 \frac{\lambda}{4} h^4 - \kappa_t \frac{y_t}{\sqrt{2}}\, \bar{t}th - \kappa_{2t} \frac{y_t}{\sqrt{2}v}\, \bar{t}th^2\,, \tag{8}$$

where κ_3 and κ_4 were introduced in Eqs. (2) and (5), respectively, and $y_t = \sqrt{2}m_t/v$, with m_t denoting the top-quark mass. The parameter κ_t thus encodes modifications of the top-quark Yukawa coupling, while the parameter κ_{2t} parametrises the coupling between two top quarks and two Higgs bosons. The SM is recovered from Eq. (8) for $\kappa_3 = \kappa_4 = \kappa_t = 1$ and $\kappa_{2t} = 0$.

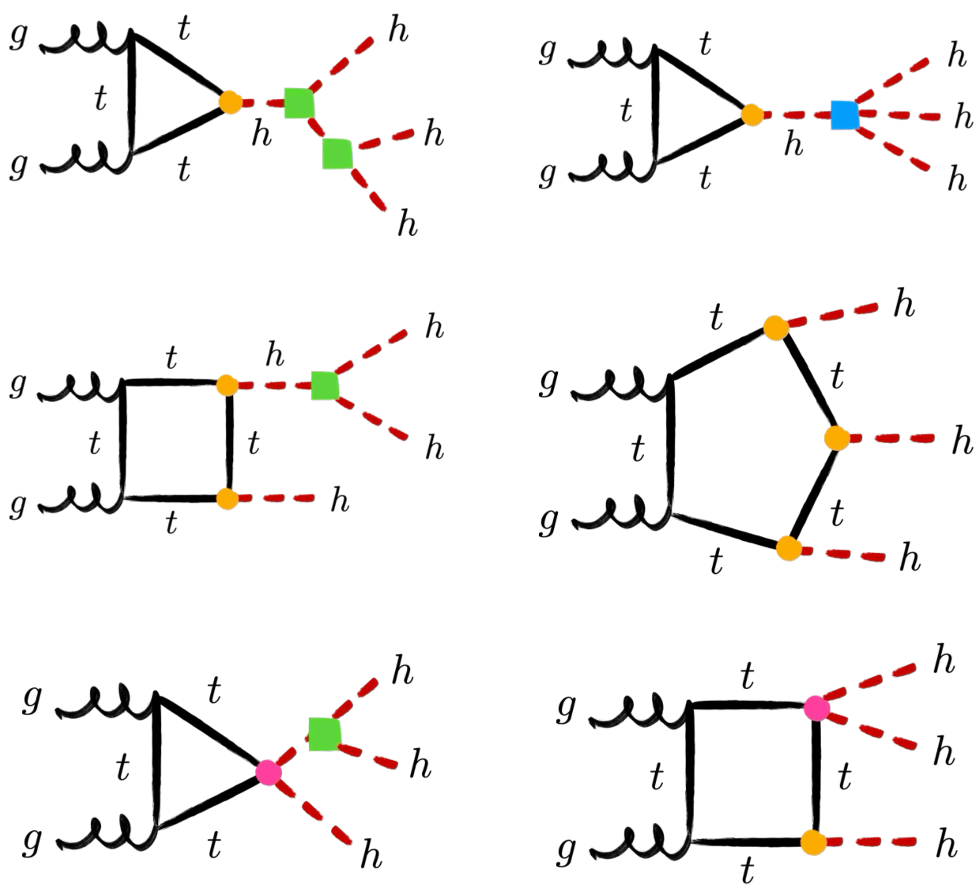

Figure 1: Representative one-loop Feynman diagrams contributing to triple-Higgs production via gluon fusion, illustrating the triangle, box, and pentagon topologies. Green (blue) boxes denote the Higgs trilinear (quartic) self-coupling, while orange (magenta) circles indicate top-quark couplings involving one (two) Higgs bosons.

To determine the dependence of the inclusive $gg \to 3h$ production cross section on κ_3, κ_4, κ_t, and κ_{2t}, we employ the leading-order (LO) matrix elements calculated in Ref. [39] and implemented in the MCFM [42–45] parton-level event generator. We use $m_h = 125\,\mathrm{GeV}$ and $m_t = 173\,\mathrm{GeV}$ as input parameters together with the PDF4LHC21 parton distribution functions (PDFs) [46]. The renormalisation and factorisation scales, μ_R and μ_F, are chosen dynamically as $\mu_R = \mu_F = m_{3h}/2$, where m_{3h} denotes the invariant mass of the three-Higgs system. For collision energies of $\sqrt{s} = 13\,\mathrm{TeV}$, $\sqrt{s} = 13.6\,\mathrm{TeV}$, and $\sqrt{s} = 14\,\mathrm{TeV}$, we parametrise the LO cross section as

$$\sigma_{\rm LO}^{\sqrt{s}} = \left(\sigma_{\rm LO}^{\sqrt{s}}\right)_{\rm SM} \sum_{n,m} c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t}) \left(\Delta\kappa_3\right)^n \left(\Delta\kappa_4\right)^m , \tag{9}$$

where $\Delta\kappa_i = \kappa_i - 1$ for $i = 3, 4$. The corresponding LO SM cross sections are

$$\sigma_{\rm LO}^{13\,\rm TeV} = 42.1\,\mathrm{ab}\,, \qquad \sigma_{\rm LO}^{13.6\,\rm TeV} = 47.9\,\mathrm{ab}\,, \qquad \sigma_{\rm LO}^{14\,\rm TeV} = 51.0\,\mathrm{ab}\,. \tag{10}$$

For $\sqrt{s} = 13\,\mathrm{TeV}$, the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eq. (9) are

$$c_{00}^{13\,\rm TeV}(\kappa_t, \kappa_{2t}) = -6.05 \cdot 10^{-4} + 0.0384\kappa_t^2 - 0.350\kappa_t^3 + 1.47\kappa_t^4 - 3.04\kappa_t^5 + 2.89\kappa_t^6$$

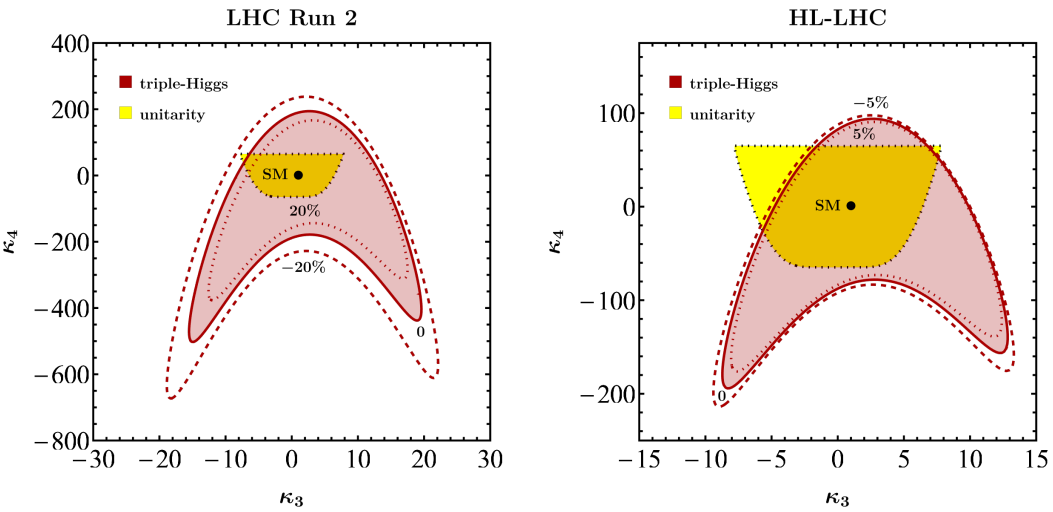

Figure 2: Constraints in the κ_3–κ_4 plane at LHC Run 2 (left) and the HL-LHC (right). The red contours indicate the preferred 95% CL regions obtained from triple-Higgs production in gluon fusion for three representative values of κ_t, as specified by the contour labels. We assume $\kappa_{2t} = 0$. The SM prediction is denoted by black points. The yellow regions, enclosed by black dotted contours, represent the perturbative unitarity bounds derived from tree-level $hh \to hh$ scattering.

$$\begin{aligned} &+\kappa_{2t}\left(1.61\kappa_t^2 - 7.82\kappa_t^3 + 15.6\kappa_t^4 - 14.8\kappa_t^5\right) \\ &+\kappa_{2t}^2\left(6.96 - 22.3\kappa_t + 37.0\kappa_t^2\right), \end{aligned} \tag{11}$$

$$c_{01}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0829\kappa_t^2 - 0.264\kappa_t^3 + 0.0895\kappa_t^4 + \kappa_{2t}\left(0.673\kappa_t - 1.12\kappa_t^2\right), \tag{12}$$

$$c_{02}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0169\kappa_t^2, \tag{13}$$

$$\begin{aligned} c_{10}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = {}& 0.211\kappa_t^2 - 1.45\kappa_t^3 + 3.65\kappa_t^4 - 3.28\kappa_t^5 \\ &+\kappa_{2t}\left(3.49\kappa_t - 13.4\kappa_t^2 + 15.6\kappa_t^3\right) + \kappa_{2t}^2(13.9 - 22.3\kappa_t) \end{aligned} \tag{14}$$

$$c_{11}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0978\kappa_t^2 - 0.264\kappa_t^3 + 0.673\kappa_{2t}\kappa_t, \tag{15}$$

$$c_{20}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.288\kappa_t^2 - 1.25\kappa_t^3 + 1.84\kappa_t^4 + \kappa_{2t}\left(2.81\kappa_t - 6.70\kappa_t^2\right) + 6.96\kappa_{2t}^2, \tag{16}$$

$$c_{21}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0489\kappa_t^2 \tag{17}$$

$$c_{30}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.159\kappa_t^2 - 0.417\kappa_t^3 + 0.938\kappa_{2t}\kappa_t, \tag{18}$$

$$c_{40}^{13\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0396\kappa_t^2. \tag{19}$$

At $\sqrt{s} = 13.6\,\mathrm{TeV}$, the corresponding coefficients read

$$\begin{aligned} c_{00}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = {}& -9.34\cdot 10^{-3} + 0.0572\kappa_t^2 - 0.446\kappa_t^3 + 1.68\kappa_t^4 - 3.17\kappa_t^5 + 2.89\kappa_t^6 \\ &+\kappa_{2t}\left(1.59\kappa_t^2 - 7.69\kappa_t^3 + 15.3\kappa_t^4 - 14.6\kappa_t^5\right) \\ &+\kappa_{2t}^2\left(6.89 - 22.1\kappa_t + 36.8\kappa_t^2\right), \end{aligned} \tag{20}$$

$$c_{01}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0814\kappa_t^2 - 0.260\kappa_t^3 + 0.0866\kappa_t^4 + \kappa_{2t}\left(0.665\kappa_t - 1.10\kappa_t^2\right), \tag{21}$$

$$c_{02}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0167\kappa_t^2, \tag{22}$$

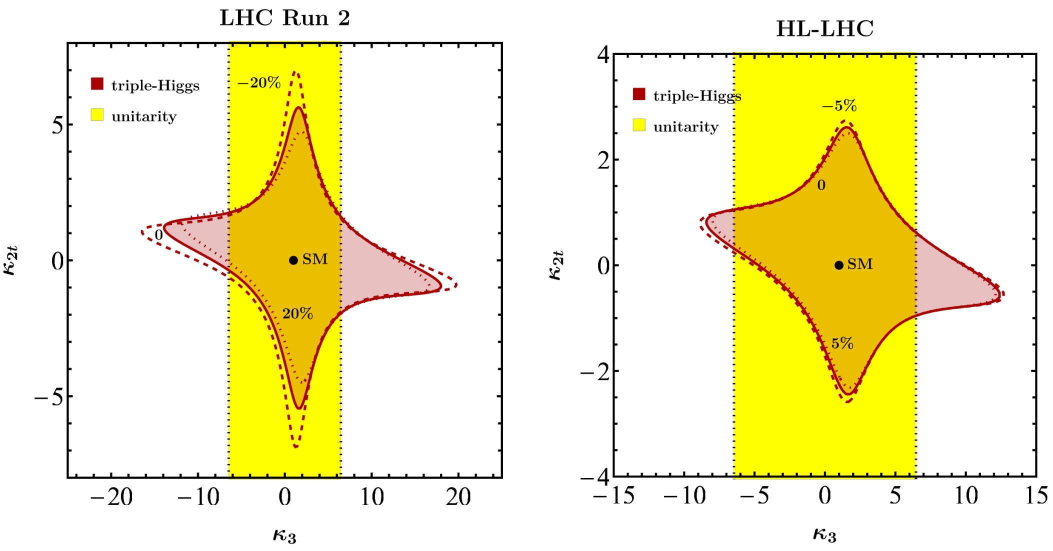

Figure 3: As Figure 2, but in the κ_3–κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). The parameters not shown explicitly are fixed to their SM values, i.e. $\kappa_4 = 1$ and $\kappa_t = 1$.

$$\begin{aligned} c_{10}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) &= 0.204\kappa_t^2 - 1.40\kappa_t^3 + 3.56\kappa_t^4 - 3.21\kappa_t^5 \\ &\quad + \kappa_{2t}\left(3.43\kappa_t - 13.2\kappa_t^2 + 15.3\kappa_t^3\right) + \kappa_{2t}^2\left(13.8 - 22.1\kappa_t\right) \end{aligned} \tag{23}$$

$$c_{11}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0961\kappa_t^2 - 0.260\kappa_t^3 + 0.665\kappa_{2t}\kappa_t\,, \tag{24}$$

$$c_{20}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.283\kappa_t^2 - 1.23\kappa_t^3 + 1.81\kappa_t^4 + \kappa_{2t}\left(2.77\kappa_t - 6.59\kappa_t^2\right) + 6.89\kappa_{2t}^2\,, \tag{25}$$

$$c_{21}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0480\kappa_t^2 \tag{26}$$

$$c_{30}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.155\kappa_t^2 - 0.408\kappa_t^3 + 0.922\kappa_{2t}\kappa_t\,, \tag{27}$$

$$c_{40}^{13.6\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0388\kappa_t^2\,. \tag{28}$$

Finally, for $\sqrt{s} = 14\,\mathrm{TeV}$, we obtain

$$\begin{aligned} c_{00}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) &= -2.18 \cdot 10^{-3} + 0.0323\kappa_t^2 - 0.311\kappa_t^3 + 1.35\kappa_t^4 - 2.92\kappa_t^5 + 2.84\kappa_t^6 \\ &\quad + \kappa_{2t}\left(1.61\kappa_t^2 - 7.79\kappa_t^3 + 15.5\kappa_t^4 - 14.7\kappa_t^5\right) \\ &\quad + \kappa_{2t}^2\left(6.99 - 22.4\kappa_t + 37.4\kappa_t^2\right)\,, \end{aligned} \tag{29}$$

$$c_{01}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0822\kappa_t^2 - 0.262\kappa_t^3 + 0.0865\kappa_t^4 + \kappa_{2t}\left(0.672\kappa_t - 1.12\kappa_t^2\right)\,, \tag{30}$$

$$c_{02}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0169\kappa_t^2\,, \tag{31}$$

$$\begin{aligned} c_{10}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) &= 0.194\kappa_t^2 - 1.39\kappa_t^3 + 3.58\kappa_t^4 - 3.24\kappa_t^5 \\ &\quad + \kappa_{2t}\left(3.46\kappa_t - 13.3\kappa_t^2 + 15.5\kappa_t^3\right) + \kappa_{2t}^2\left(14.0 - 22.4\kappa_t\right) \end{aligned} \tag{32}$$

$$c_{11}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0968\kappa_t^2 - 0.262\kappa_t^3 + 0.672\kappa_{2t}\kappa_t\,, \tag{33}$$

$$c_{20}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.284\kappa_t^2 - 1.23\kappa_t^3 + 1.82\kappa_t^4 + \kappa_{2t}\left(2.79\kappa_t - 6.65\kappa_t^2\right) + 6.99\kappa_{2t}^2\,, \tag{34}$$

$$c_{21}^{14\,\mathrm{TeV}}(\kappa_t, \kappa_{2t}) = 0.0484\kappa_t^2 \tag{35}$$

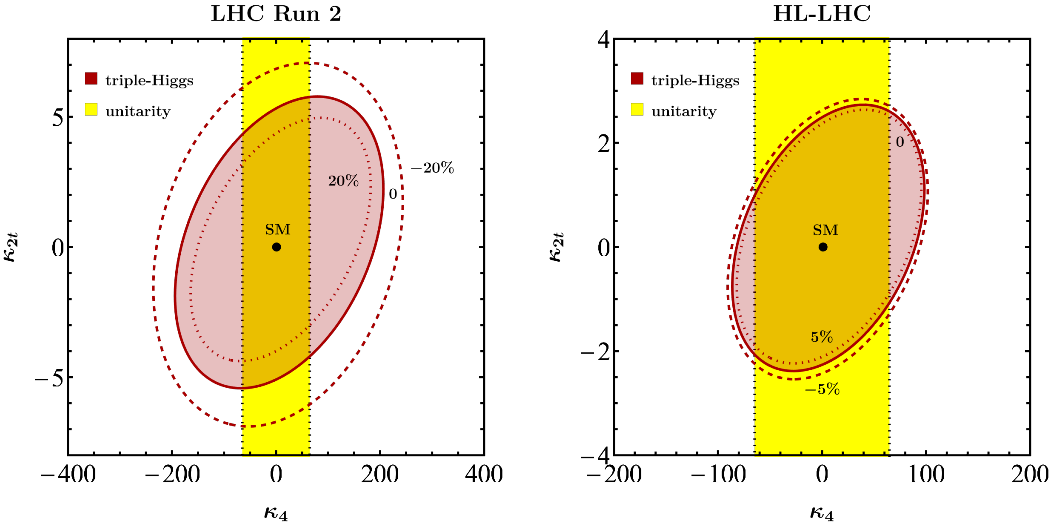

Figure 4: As Figure 2, but in the κ_4–κ_{2t} plane at LHC Run 2 (left) and the HL-LHC (right). The remaining parameters are fixed to their SM values, i.e. $\kappa_3 = 1$ and $\kappa_t = 1$.

$$c_{30}^{14\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.156\kappa_t^2 - 0.411\kappa_t^3 + 0.931\kappa_{2t}\kappa_t\,, \tag{36}$$

$$c_{40}^{14\,\text{TeV}}(\kappa_t, \kappa_{2t}) = 0.0391\kappa_t^2\,. \tag{37}$$

Notice that the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ depend only mildly on the collision energy over the range considered. Likewise, the choice of PDFs is expected to have a negligible numerical impact on the parameterisations given in Eqs. (11) to (37). Higher-order QCD corrections to the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ are currently unknown, although they could, in principle, be estimated in the infinite top-quark mass limit. In the SM, these corrections increase the inclusive cross section by a K-factor of approximately 1.7 at NNLO in QCD [18]. Parameterisations of the LHC triple-Higgs gluon-fusion production cross section similar to Eqs. (9) to (37) have also been presented in Refs. [35, 36, 38, 39].

The two panels in Figure 2 illustrate the expected sensitivity to the Higgs self-couplings κ_3 and κ_4 using current LHC Run 2 data and HL-LHC projections, assuming an integrated luminosity of $3\,\text{ab}^{-1}$. The red contours indicate the 95% CL constraints derived from triple-Higgs production in gluon fusion using the parameterisations in Eqs. (11) to (19) and Eqs. (29) to (37), with the normalisation to the corresponding LO SM cross section given in Eq. (10). For LHC Run 2, the triple-Higgs constraints are obtained by imposing $\mu_{3h}^{\text{LHC Run 2}} < 588$ [20], while for the HL-LHC we assume a projected upper limit of $\mu_{3h}^{\text{HL-LHC}} < 125$ [47]. Throughout this analysis, we set $\kappa_{2t} = 0$, and the black dots indicate the SM prediction. The left panel demonstrates that modifications of the top-quark Yukawa coupling of $\Delta\kappa_t = \pm 20\%$, where $\Delta\kappa_t = \kappa_t - 1$, which remain allowed at the 95% CL after LHC Run 2 [48, 49], have a significant impact on the resulting constraints. This behaviour was already pointed out in Ref. [41] and originates from the strong power-law dependence of the coefficients $c_{nm}^{\sqrt{s}}(\kappa_t, \kappa_{2t})$ appearing in Eqs. (11) to (37) on κ_t. By the end of the HL-LHC, the top-quark Yukawa coupling is expected to be measured with a precision of $\Delta\kappa_t = \pm 5\%$ at the 95% CL [50]. Consequently, its impact on the $gg \to 3h$ constraints in the κ_3–κ_4 plane will be substantially reduced, as illustrated in the right panel. Finally, we note that the projected HL-LHC constraints from $gg \to 3h$ are comparable to those imposed by perturbative unitarity [34, 51, 52], shown as the yellow region enclosed by the black-dotted contour in both panels.

The panels in Figures 3 and 4 show the expected sensitivity in the κ_3–κ_{2t} and κ_4–κ_{2t}

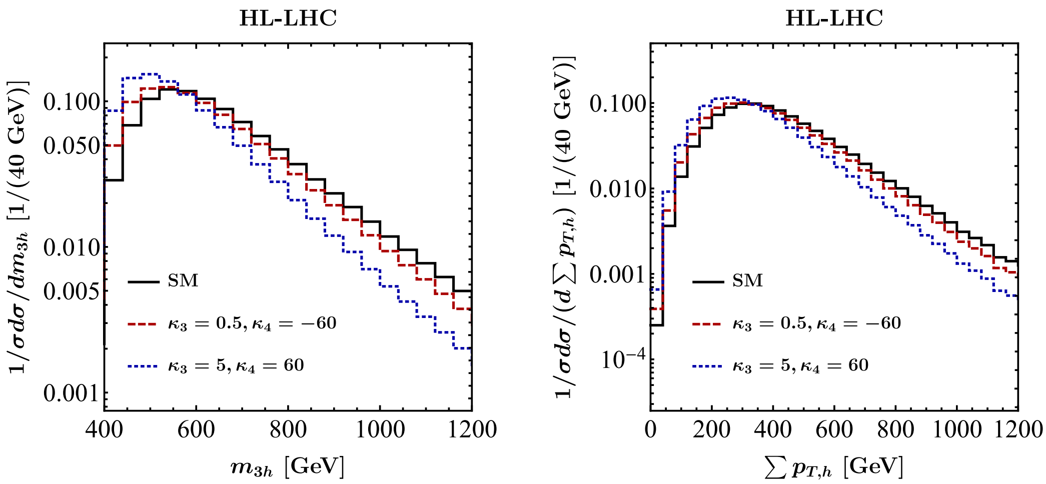

Figure 5: Normalised kinematic distributions for $gg \to 3h$ production at the HL-LHC in the SM (solid black line) and in two benchmark scenarios defined by κ_3 and κ_4 (red dashed and blue dotted lines). The left and right panels show the invariant mass of the three-Higgs system, m_{3h}, and the scalar sum of the transverse momenta of the Higgs bosons, $\sum p_{T,h}$, respectively. All results are obtained for $\kappa_t = 1$ and $\kappa_{2t} = 0$.

planes, respectively. The red contours represent the 95% CL constraints from triple-Higgs production in gluon fusion, obtained following the procedure described above. Parameters not shown explicitly are fixed to their SM values, and the black dots indicate the SM prediction. As in Figure 2, the left panels demonstrate that variations of the top-quark Yukawa coupling of $\Delta\kappa_t = \pm 20\%$ can have a significant impact on the resulting constraints, whereas the smaller variations $\Delta\kappa_t = \pm 5\%$ considered in the right panels lead to only minor changes. For κ_{2t}, triple-Higgs production at LHC Run 2 yields $-5.1 < \kappa_{2t} < 5.3$, while the HL-LHC is expected to improve this range to $-2.3 < \kappa_{2t} < 2.5$ at 95% CL. These limits are obtained by fixing $\kappa_3 = \kappa_4 = \kappa_t = 1$. We note that, under the same assumptions, the 95% CL constraint on the double-Higgs production signal strength obtained at LHC Run 2, $\mu_{2h}^{\text{LHC Run 2}} < 2.5$ [17], already implies $-0.19 < \kappa_{2t} < 0.72$. The quoted bound is obtained using the expressions given in Refs. [53, 54]. Consequently, triple-Higgs production is not expected to provide a stronger constraint on κ_{2t} than double-Higgs production in a single-parameter analysis. It can, however, provide complementary information in multi-parameter fits, in particular when these involve κ_4, which can only be weakly constrained by double-Higgs production [21–24].

Modifications of the Higgs trilinear and quartic self-couplings affect not only the inclusive triple-Higgs production cross section but also the kinematic properties of the $gg \to 3h$ final state. To illustrate this point, Figure 5 shows normalised kinematic distributions at the HL-LHC for the SM and two BSM benchmark scenarios. Both benchmark points yield the same inclusive signal strength, $\mu_{3h} = 65$, with inclusive cross sections differing by only about 5%, while remaining consistent with the perturbative unitarity bounds obtained from tree-level $hh \to hh$ scattering. The left and right panels show the invariant mass of the three-Higgs system, m_{3h}, and the scalar sum of the Higgs transverse momenta, $\sum p_{T,h}$, respectively. All results are obtained for $\kappa_t = 1$ and $\kappa_{2t} = 0$. This choice is motivated by the observation of Ref. [41] that variations of κ_t approximately induce an overall rescaling of the normalised distributions, without significantly altering their shapes, and this has been numerically verified also for $\kappa_{3,4} \neq 1$ within the unitarity bounds. The comparison demonstrates that values of κ_3 and κ_4 yielding nearly degenerate inclusive cross sections can nevertheless lead to sizeable distortions of the differential spectra, reaching up to about 50% in both the m_{3h} and $\sum p_{T,h}$ distributions,

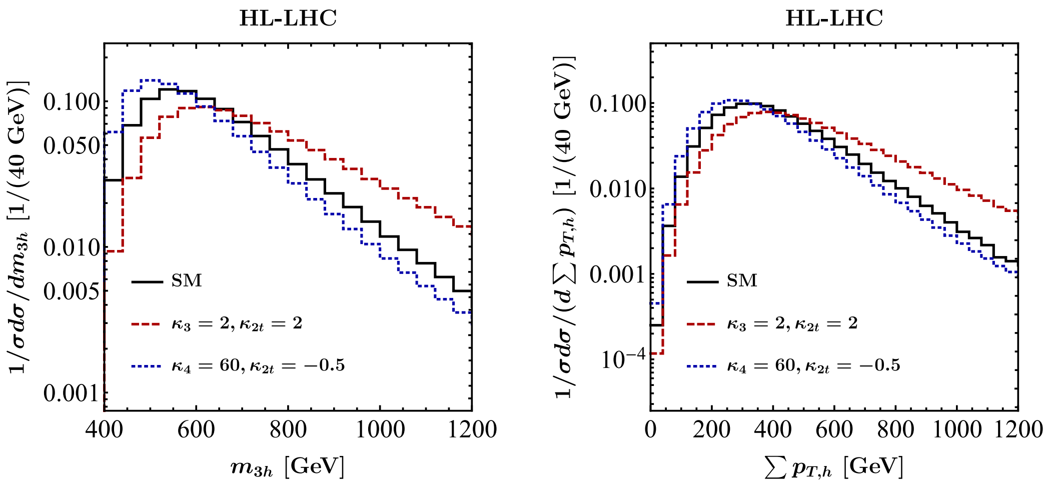

Figure 6: As in Figure 5, but for two other κ-benchmark scenarios indicated in the panels. The remaining parameters are fixed to their SM values.

in the peak regions as well as in the high-energy tails. Consequently, exploiting differential information in future HL-LHC analyses of $gg \to 3h$ production has the potential to improve the constraint quoted below Eq. (6), which currently relies on the inclusive rate only [21].

Figure 6 presents additional normalised kinematic distributions at the HL-LHC for the SM and two further BSM benchmark scenarios within the κ-framework. The benchmark points are selected such that both lead to an enhanced signal strength of $\mu_{3h} = 75$, while their inclusive cross sections differ by less than about 5%. The corresponding values of κ_3 and κ_4 are consistent with perturbative unitarity constraints, and all remaining parameters are fixed to their SM values. As in Figure 5, sizeable distortions of the differential spectra arise despite the near-degeneracy of the inclusive rates. This further demonstrates that exploiting differential information in future HL-LHC searches for $gg \to 3h$ production can help lift parameter degeneracies within the κ-framework.

It is worth noticing that if the higher-order operator $\frac{m_t}{v^3}\frac{\kappa_{3t}}{2} t\bar{t}h^3$ is introduced [35], similar conclusions hold. Scanning benchmark points with $\kappa_{3,4} \neq 1$ and $|\kappa_{3t}| \lesssim 5$, a range still experimentally allowed, generically shifts the inclusive $gg \to 3h$ cross section far from its SM prediction, leaving little room for a dedicated shape analysis. For $\kappa_{3,4} \sim 1$ and $\kappa_{3t} \lesssim 0.5$, however, the cross section remains close to the SM prediction while the distributions quickly become visibly distorted, indicating that, given sufficient integrated luminosity, shape information could in principle help discriminate such scenarios from the SM.

3 Conclusions

Unlike the Higgs couplings to EW gauge bosons and third-generation fermions, which are already being measured with increasing precision, the Higgs self-interactions remain largely unconstrained by LHC Run 2 data. This leaves considerable room for BSM scenarios that modify the shape of the Higgs potential. Indeed, both theoretical considerations and explicit ultraviolet-complete constructions demonstrate that deviations of the Higgs self-couplings exceeding 200% can be consistent with existing single-Higgs measurements without requiring fine-tuned parameter choices [4]. Measurements of multi-Higgs production at the HL-LHC and future colliders therefore provide a unique opportunity to probe largely unexplored regions of parameter space and retain significant discovery potential, even in scenarios where single-Higgs observables remain close to their SM predictions.

In this note, we have derived compact "pocket formulas" for triple-Higgs production within the κ-framework, enabling constraints on the Higgs trilinear and quartic self-couplings to be obtained from current and future measurements of the $gg \to 3h$ signal strength. The resulting parameterisations retain the full dependence on κ_3, κ_4, κ_t, and κ_{2t} at LO in perturbation theory. Since their derivation relies on the flexible and efficient matrix-element implementation developed in Ref. [39], the extension of these formulas to incorporate additional coupling modifications — such as those arising at next-to-leading order in the Higgs Effective Field Theory [54] — is comparatively straightforward.

Using these parameterisations, we have investigated the sensitivity to κ_3 and κ_4 using current and projected LHC data. We find that the presently allowed uncertainty in κ_t can have a substantial impact on the constraints in the κ_3–κ_4 plane. This dependence is significantly reduced at the HL-LHC, where the expected percent-level precision of the top-quark Yukawa coupling measurement will strongly restrict the corresponding parameter space. For κ_{2t}, which parametrises the interaction between two top quarks and two Higgs bosons, we find that triple-Higgs production is unlikely to provide stronger bounds than double-Higgs production in a single-parameter analysis. Nevertheless, it can yield complementary information in global multi-parameter fits, especially once κ_4 is included, as this coupling remains only weakly constrained by double-Higgs measurements [21–24]. We further demonstrate that parameter choices for κ_3 and κ_4 yielding nearly identical inclusive $gg \to 3h$ cross sections can still produce substantial differences in differential observables. Similar effects are observed for variations of κ_{2t}.

The projected HL-LHC sensitivity to κ_4 in a single-parameter analysis remains modest, with constraints reaching only $|\kappa_4| \lesssim 80$, comparable to those derived from perturbative unitarity arguments [34, 51, 52]. An improvement by approximately one order of magnitude in the determination of the Higgs quartic self-coupling could be achieved at a hadron-collider option of the Future Circular Collider [22, 28–30, 32, 33, 35–38, 40]. Reaching a precision at the level of 50%, however, would likely require the capabilities of a future muon collider [55].

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